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Prevalence and trends of hepatitis B and C virus biomarkers in Zimbabwe: comparative analyses of a nation's blood-donor surveillance data and meta-analyses of population studies

Simbarashe Mabaya[a], **Edmore Munongo**[a], **Tonderai Mapako**[b], **Lucy Marowa**[b], **Alex N. Gasasira**[a], **Jotam G. Pasipanodya**[c,*], **Menard Mutenherwa**[b,*]

[a]World Health Organization (WHO) Country Office, Harare, Zimbabwe

[b]National Blood Service Zimbabwe, Harare, Zimbabwe

[c]Department of Medicine, Division of Infectious Diseases/Division of Clinical Pharmacology, Vanderbilt University Medical Center, Nashville, TN, USA

Abstract

Background: The disproportionate burden of viral hepatitis, particularly hepatitis B virus (HBV) is experienced by people living in low-resourced sub-Saharan Africa, where the estimated prevalence is 3–7 times the global average. Therefore to inform policy, we describe the seroprevalence and trends of hepatitis C (HCV) and HBV biomarkers: anti-HCV antibody and hepatitis B surface antigen (HBsAg), respectively, in Zimbabwe.

Methods: We analysed data from 181,248 consecutive blood-donors, examined between January 2015 through December 2018. Additionally, we conducted a comprehensive literature review using PubMed and African Journals Online databases, meta-analysing selected papers from Zimbabwe, published between 1970 and 2020, that met specific criteria.

Results: Overall age-standardized prevalence rate (ASPR) for anti-HCV was 8.67 (95%CI, 0.25–17.09) per 100,000, while that for HBsAg was 2.26 (95%, 1.89–2.63) per 1000 blood-donors, per year. Meta-analysis of 9 studies comprising 220,127 persons tested for anti-HCV revealed ASPR of 0.05% (95% 0%–0.19%) in blood-donors and 1.78% (95%CI, 0.01%–5.55%) in the

CONTACT: Jotam G. Pasipanodya, jpasipanodya@hotmail.com, Department of Medicine, Division of Infectious Diseases/ Division of Clinical Pharmacology, Vanderbilt University Medical Center, Nashville, TN, USA, Menard Mutenherwa, mmutenherwa@nbsz.co.zw, National Blood Service Zimbabwe (NBSZ). Box A101, Avondale, Harare, Zimbabwe.
*These authors contributed equally to this work.

Authors contributions

SM wrote the first draft; SM, EM, MM and JGP conceptualised and designed the study; SM, MM and JGP performed the systematic review; JGP performed the meta-analysis; SM and JGP analysed data; SM, EM, MM, TM, LM, MM and JGP reviewed the results and verified the underlying data. All authors contributed to the final manuscript and approved the final version.

Disclosure statement

The authors have no competing interests to declare. Some authors were paid salaries by the WHO; however, they did not receive specific funding for this study. Some data in this study are collected and managed by NBSZ; however, the authors were responsible for study design, analysis, interpretation, and writing of this Article. The corresponding author (MM) had full access to all the data and had final responsibility for the decision to submit for publication

Ethical approval

Ethical approval was not required for the meta-analysis. However, for the observational retrospective analysis of the blood donors data, ethical approval was obtained from NBSZ (NBSZ006/2019) and Medical Research Council of Zimbabwe (MRCZ/E/261).

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general population, for an overall pooled ASPR of 0.44 (95%CI, 0.19%–0.76%). 21 studies comprising 291,784 persons tested for HBsAg revealed ASPR of 0.65% (95%CI, 0.31%–1.00%) in blood-donors and 4.31% (95%CI, 1.77%–6.50%) in the general population for an overall pooled ASPR of 4.02% (95%CI, 3.55%–4.48%), after HBV vaccine introduction. HBsAg prevalence was significantly higher before HBV vaccine introductions.

Conclusions: The prevalence of HBV is decreasing, consistent with the introduction of HBV vaccination, while HCV prevalence is increasing in Zimbabwe. This highlights the need for Improved blood-donor screening and more informative biomarker studies, particularly among repeat donors and children.

Keywords

Hepatitis B vaccine; age-standardized prevalence; crude prevalence; anti-hepatitis C virus; hepatitis B surface antigen; Zimbabwe blood transfusion screening; hepatitis B virus/hepatitis C virus co-infection

Background

There has been remarkable progress with point-of-care diagnostics, blood-donor screening algorithms for transfusion transmittable infections (TTI), and chemotherapeutics, especially with directly acting antiviral agents (DAA) [1–5]. This makes attaining World Health Organization (WHO) global targets to eliminate hepatitis infections by year 2030 feasible. Unfortunately, the disproportionate burden due hepatitis C virus (HCV) as well as hepatitis B virus (HBV), is experienced by people living in low-resourced sub-Saharan Africa, where the estimated HBV prevalence is 3–7 times the global average [6,7]. The WHO estimate about 1.6 million chronic HBV infections that contribute about 2500 deaths each year, however, there are no official national estimates on HCV burden, with/out HIV coinfection [8]. Another equally troubling fact is that up-to-date epidemiology data needed for risk-based decision-making to contain these viral infections, including crafting treatment guidelines, is not readily available in those low-resources setting, such as Zimbabwe [7,9–11]. Therefore, it is unclear whether uniform public health interventions and targets set by WHO and recommended by experts will succeed when applied across these different settings.

Zimbabwe is a country undergoing epidemiological and health transition, with several vulnerable population groups attended to by fragile health care systems. This impacts both the diagnostic testing and surveillance of general diseases. Therefore, the introduction of HBV vaccination by the Zimbabwean government in the late 1990s *via* an existing expanded programme on immunisation (EPI) in infants, was an inexpensive and economically attractive means to attain universal vaccination for birth cohorts without use of serological screening. Similarly, routine screening of blood-donations for TTI, including HCV and HBV, can be regarded as cost-effective active surveillance for those two viral infections in low-risk healthy population groups. Indeed, anti-HCV antibody and HBsAg test are widely used as biomarkers for HCV and HBV in the clinic in Zimbabwe. However, the contextual utility of those two biomarkers as population screening tools is unknown.

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Our study objective was to estimate prevalence and describe the burden of viral hepatitis among routine blood-donors and the general population in Zimbabwe using the widely used anti-HCV antibody and HBsAg biomarkers [12–16].

Methods

We undertook three different analyses to examine HBV and HCV prevalence in Zimbabwe. First, we performed a HBV and HCV biomarker prevalence study in a cohort of national blood-donors, and then undertook systematic reviews and a meta-analysis of seroprevalence studies for the same biomarkers from Zimbabwe, from 1970 through 2020. Third, we computed ratio of HBV and HCV prevalence in blood-donors compared to the general public from the latter.

HBV and HCV seroprevalence in Zimbabwean blood-donors'

We conducted a retrospective study utilising routinely collected data from January 2015 to December 2018, from the National Blood Service Zimbabwe (NBSZ). All data were de-identified and the links to personal identification data of the donors were not available to the investigators. Ethical approval for this study was obtained from NBSZ (NBSZ006/2019) and Medical Research Council of Zimbabwe (MRCZ/E/261). Blood-donors were drawn from the 5 major cities in Zimbabwe, including major administrative NBSZ districts: Harare, Bulawayo, Gweru, Mutare and Masvingo. Together, these geographic locations represent an estimated 15 million inhabitants in Southern Africa.

Study setting

Small surveys and observational studies done in selected populations have reported inconsistently low HCV prevalence [16–19], while several population surveys suggests that HBV is endemic in Zimbabwe [6,7,13,15,16,20,21]. Country-wide universal early childhood monovalent HBV vaccinations was introduced in 1994, and the 3-dose pentavalent (DTwPHibHepB) vaccine was introduced in 2000, while timely administration of birth-dose HBV vaccine is scheduled to start in 2021 or 2022. The vaccination coverage for the 3-doses has been persistently above 90% ever since [7]. Further country details on HBV prevalence, including studies of pregnant women [17,20,22,23], are given in the given in the webappendix online and meta-analysis formerly described below.

NBSZ is a public-private partnership organisation that has national mandate for the collection, screening and distribution of donated blood and blood products in the country for the past 50 years [9]. All blood donations in Zimbabwe are 100% voluntary and non-remunerated donations obtained from few selective low-risk populations. Persons with a history of high-risk behaviours, including intravenous drug use (IVDU) or transactional sexual practices are precluded from blood donations, based on all-inclusive pre-screening oral questionnaire and physical examinations. Only persons of 16–75 years of age may become blood-donors.

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Laboratory procedures and data sources

NBSZ procedures and protocols for screening eligible blood donations for the four common TTI: HBV, HCV, Human immunodeficiency virus (HIV) and Treponema pallidum, have been reported extensively [9,11,24,25]. Briefly, serum from each donation is initially tested for all four TTI using Abbott Architect and related reagents, and any non-reactive donation is declared free of TTI. Reactive samples undergo two simultaneous tests: one with the same Abbott Architect, and the second, using different technology. The same testing algorithm is used for all TTI with minor variations for the secondary test. At the moment, NBSZ does not use qualitative or quantitative tests to ascertain HCV RNA or HBV DNA positivity in mini-pools or individual donations.

Data were extracted from the NBSZ Blood Bank Management System (e-Delphyn) for analyses. Donor information is captured through paper-based donor enrolment and assessment form and entered into e-Delphyn. The donor enrolment and assessment form also serves as a risk assessment tool and screens for high-risk behaviours in potential donors. Each donor is assigned a unique identifier for life, however, only aggregated data were extracted for analyses.

Statistical analysis

We calculated the crude prevalence of HBsAg per 1000 and anti-HCV per 100,000 donor populations, and their respective exact 95% confidence intervals [CI], in STATA version 16 (College Station, Texas) and GraphPad (San Diego, California) software. City-specific, calendar-year-specific and age-group-cohort-specific crude prevalence were calculated to estimate the geographic, period and cohort disease burden, respectively. To facilitate cross comparison of prevalence estimates obtained we used WHO World Population Standards and approach to obtain age-standardized prevalence rates (ASPR) [26]. ASPR and 95%CI were calculated using the WHO standard population weights, which was constructed to reflect the average world population age-structure for the period 2000–2025 [26,27]. This approach allows for less biased epidemiological comparison across countries in time and between groups [27]. Next, to correlate viral burden in blood-donors to that of the general Zimbabwean public, we computed transportable ratio estimates that could be used for surveillance purposes. The general population (GP) to blood-donor (BD) prevalence ratio and corresponding confidence interval is introduced as a marker to track dual temporal change in HBsAg prevalences.

Lastly, in order to inform policy, we searched the entire body of literature on HBV and HCV prevalence in Zimbabwe, specifically tested for temporal trends in blood-donors, the general public and notable risk groups, to determine whether HBV vaccination impacted those HBV prevalence trends [1,2,12].

Systematic review and meta-analysis of HBV and HCV seroprevalence studies

To estimate and compare HBV prevalence before and after HBV vaccination we undertook systematic review and meta-analyses of population studies and surveys of HBV performed in Zimbabwe. Similar analysis was performed for HCV, using approach and methods we

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have used in the past [28,29]. We followed the PRISMA statement (detailed in webappendix online page 2, Table S1) to search PubMed and African Journals Online databases [30].

Results

Demographic characteristics of blood donors

A total of 181,248 blood donors, 98,615 (54%) men and 82,633 (46%) women, were screened for anti-HCV and HBsAg biomarkers from January 2015 through December 2018. The majority (68%) of these blood-donors were in the 16–20 years age-group (webappendix page 7, Table S2).

Seroprevalence and temporal trend analysis of anti-HCV in blood-donors

There were 22 blood-donors seropositive for HCV, for an overall percent prevalence of 0.0012% (95% CI, 0.008%–0.018%). Of these 22, two had concurrent positive tests for HCV and syphilis, one was positive for both HCV and HIV, and none were positives for HCV and HBV. Table 1 shows that the overall anti-HCV ASPR was 8.67 (95%CI, 0.25–17.09) per 100,000 donors. Both the ASPR and crude prevalence rate (CPR) varied by geographic location, however, there was no significant gender differences (webappendix pages 2–3).

Anti-HCV ASPR significantly increased by 2.94 fold (95% CI, 2.68–3.23) $p = .002$, between 2015–2016 and 2017–2018 periods, driven mainly by increases in the >41 years age-group despite significant gradual fall in the 21–30 years age-group (Figure 1(D)). These data suggest low but unignorable increase of HCV prevalence among blood-donors.

Seroprevalence and temporal trend analysis of HBsAg in blood-donors

There were 593 blood donors seropositive for HBsAg, for an overall percent prevalence of 0.33% (95% CI, 0.30%–0.36%) and an ASPR of 2.26 (95%, 1.89–2.63) per 1000 donors. HBsAg prevalence was about two-folds significantly higher in males than females, regardless of the estimating method (Table 2). There was significant reduction in HBsAg biomarker with time, for both CPR and ASPR, with the highest CPR of 9.8 (95%CI, 7.6–12.3) per 1000 donors recorded in 2015 in Bulawayo, and the lowest of 1.02 (95%, 0.33–2.39) per 1000 donors recorded in Gweru in 2016 (Figure 2). Interestingly, both CPR and ASPR of HBsAg for the 16–20 age-groups significantly declined, while the ASPR for >41 age group, significantly increased (Figure 2).

Next, in order to characterise trends, we fitted nonlinear regression models to the data depicted in Figure 2(D), and then estimated the slope parameter for each age-group using a least-square approach. Decline in HBsAg prevalence in the 16–20 year age-group was best described by the equation: $y = -0.083\text{year} + 168$, $R^2 = 0988$. The distribution of indeterminate HBsAg test results across age-groups and geographic locations was virtually identical, suggesting that any HBsAg test or testing algorithmic error, was invariant to those two groups.

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Systematic review and meta-analyses of seroprevalence studies from Zimbabwe

Our search returned 361 unique records, of which 51 met the inclusion criteria for full review, while 31 were excluded for reasons shown in the PRSIMA Figure 3. An additional three studies were identified from references during screening. Of the 23 (45%) studies selected, only two (9%) studies reported exclusively on HCV [17,19], while another seven (30%) studies [9,16,19,24,31–34] reported on both HBV and HCV prevalences in combined 220,127 persons, shown in Table 3. Majority, 216,295 (98%) were blood-donors, 1591 (0.72%) were pregnant women and the remainder were drawn from the general population. Table 4 summarises the 21 studies [13–15,20–23,33–40] that reported HBV prevalence in 291,784 (95%) blood-donors, 1701 (06%) pregnant women and 13,658 (4.4%) drawn from the general public or population.

Only three (13%) studies [16,21,39] were considered at moderate risk of bias. Even though all studies used secondary confirmatory assays (mostly the exact same one), only three studies [32,37,39] used viral genome (HBV DNA or HCV RNA) to confirm viraemia. Thus, overall quality of selected studies was modest. There was 90% agreement on study selection and 100% in studies' quality rating between the two authors who formally performed the meta-analysis (JGP and SM). We did not identify dedicated paediatric studies, or studies that examined perinatal transmission of either virus, or commonly reported HBV and/or HCV risk factors, such as IVDU or men-who-have-sex-with-men (MSM) [2,12,41–45]. The exception was one study [38] which reported heterosexual prostitution. Therefore, the meta-analysis could be biased, especially against children, IVDU and other risk factors commonly associated with HCV or HBV [1,2].

Seroprevalence of anti-HCV in general population

The estimated overall pooled anti-HCV seroprevalence from the nine studies [9,16,17,19,24,31–34] was an ASPR of 0.44% (95%CI, 0.19%–0.76%). However, there was significant overall heterogeneity between four risk groups examined, I^2=9654%, $p < .001$. The pooled ASPR estimate, measures of heterogeneity within and between the different risk groups and bias is shown in Figure 4 and webappendix page 5, eFigure 1. The pooled anti-HCV ASPR in the general population of 1.78% (95%CI, 0.01%–5.55%) was significantly higher, $p = .019$, compared to 0.05% (95%CI, 0%–0.19%) observed in blood-donors. Sensitivity analysis revealed robust pooled ASPR estimates that only significantly varied when the 2000/2004 blood-donor dataset was removed (eFigure 2). This is because there was significant anti-HCV prevalence increase in Zimbabwe, even among blood-donors between 2000 and 2010 (Figure 4) [33]. Interestingly, that temporal prevalence increase was not apparent when examined with meta-regression (eFigure 3). Therefore, we formally tested the hypothesis that anti-HCV prevalence could be zero in the different risk groups. That hypothesis was significantly rejected in the general population ($p < .038$), in pregnant women ($p < .001$), in HIV-infected persons ($p < .001$), and overall ($p = .001$), but not in blood-donors ($p = .118$). There was no significant publication bias, 4.02 (95%CI, −0.85–8.89), and no small-study effect $p = .095$, which was unexpected based on prior systematic review described above (webappendix page 5–6, eFigures 1–3).

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Seroprevalence of HBsAg in general population

The pooled HBsAg prevalence were significantly heterogeneous with notable temporal trend differences, especially between the general population and blood-donors, in successive decades, both before and after HBV vaccine introduction periods. Figure 5 shows temporal changes for each decade, from 1970 through 2020, of pooled HBsAg ASPR measured in percentage (%). First, pooled HBsAg ASPR peaked between 2000 and 2010 at 8.15% (95% 1.35%–14.95%), which was four-times higher than those observed between 1970 and 1980. Second, in the 1970s, the mean HBsAg prevalence in blood-donors was not significantly different from that in the general population, 2.42% versus 2.16%, respectively, $p = .602$ [46]; however, this was not true in studies performed after HBV vaccine introduction, in the 2000s [9,33]. The forest plots in eFigures 5 show that pooled HBsAg prevalence among blood-donors was six-folds times significantly lower than contemporary studies in the general population, 12-times lower than in pregnant women and 14-times lower than that observed in HIV-infected patients. Overall estimated pooled ASPR HBsAg in the post HBV vaccination period was 4.02% (95%CI, 3.55%–4.48%), which was significantly lower than estimates prior to HBV vaccination. The GP/BD HBsAg prevalence ratio after HBV introduction was 6.35 (95%CI, 6.32–6.39) compared to 2.47 (95%CI, 2.45–2.49) before HBV introduction. Finally, five studies [15,20,22,24,37] measured HBeAg in 1,152 persons, and these revealed a pooled positivity rate of 20% (95%, 9%–34%) (eFigure 6). However, there was significant publication bias, 7.97 (95%CI, 3.87–12.08), with statistically significant small study effect ($p = .014$) (eFigure 8).

Next, we used meta-regression to identify study-level factors associated with change in effect size. Table 5 shows that on average HBsAg ASPR decreased by 2.45% with each decade if we hold general population risk factors constant. In addition, HBsAg prevalence was higher in studies of HIV-infected and/or pregnant women than studies of general population by a mean of 7.80%, suggesting that higher HBsAg prevalence in at-risk-population was a persistent feature after HBV vaccine introduction.

Discussion

We found low prevalence of viral hepatitis infections in Zimbabwean blood-donors since 180,423 (99.5%) out of 181,248 unique donors were non-reactive for either anti-HCV or HBsAg for the study period. Apata et al. recently reported summaries of HBV- and HCV-related surveillance data for 38 sub-Saharan African countries for the years 2000 through 2011, which is routinely captured by the WHO's Global Database on blood safety [33,47]. The CPR of HCV in Zimbabwe increased from 0.03% in 2000/2004 to 0.34% in 2010/2011, while HBV decreased for the same period from 1.56% to 0.92%, respectively. In our study, for the period 2015 through 2018 we observed CPR of 0.0012% for HCV and 0.33% for HBV. While the general decrease in HBV prevalence in blood-donor follows a consistent declining pattern [9,40], the low HCV prevalence observed is unrealistic, and not consistent with patterns and contemporary estimates observed in the general population [31,34]. First, Moyo et al. reported anti-HCV CPR of 5.7% in men and 11.4% in women, consistent with earlier reports by Gangaidzo and others [16,34]. Moreover, both studies reported elevated hepatic transaminase in those affected suggesting ongoing viremia [42,44]. Second, there

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were high rate of indeterminate anti-HCV results with discordant patterns when compared to positive results suggesting systemic bias (Table 1). Third, the rate and factors associated with spontaneous HCV clearance with our study populations is unknown [42,48]. However, observational studies suggests that the variances would be larger with sub-Saharan Africans suggesting that, with observed low HCV prevalence, both high false positive and false negative results would be expected with the current assays and testing algorithm used by NBSZ [1,42]. There are several other reasons for false results with the HCV serology tests used in study, including prolonged window period from viremia before seroconversion and detection of anti-HCV. Benchmarking blood-donor screening tests and protocols to align with best practices used in the developed world warrants consideration [5]; not only because it would significantly reduce high residual risk for HCV [49], but also improve our understanding of HCV epidemiology by notifying each blood-donors of their true HCV status [1,2,5,10,45,49]. The current practice is that blood-donors with indeterminate results are automatically dropped as regular donors, after brief counselling.

Indeed, individual HIV, HCV and HBV can exacerbate pathogenesis of the other, and they each share common transmission routes in our clinical settings. The frequency with which each of those two most common transmission routes (transfusion versus vertical) contribute to viral hepatitis disease burden in Zimbabwe is unclear. However, one study [31] in our meta-analysis attributed all 2 HCV cases in that study to prior blood transfusion. Compared to one in 2 million donations reported and considered acceptable in the US [50], at risk rates of 2.5 for HCV and 4.3 for HBV per 1000 donations with current NBSZ's screening and testing algorithms, blood transfusions still pose significantly increased risk in Zimbabwe [33,49,50]. This is particularly concerning for complicated pregnancy during malaria outbreaks when frequent blood transfusions are given to women, or for haematological malignancy in children requiring multiple large volume transfusions [17,20,22,23,40].

None of the blood-donors in our study cohort was positive for HBV/HCV co-infection. Given that there are currently no clear data to direct optimal management of HBV/HCV coinfections [1], and initiating DAA for HCV has been associated with recurrence of current or previously suppressed HBV coinfection resulting in serious liver problems or death [51], this was a reassuring finding and indicator. These data confirm that HBV prevalence is high in Zimbabwe. Furthermore, our review of the literature revealed that the demographic composition of the Zimbabwean blood-donors as well as their risk profiles for viral hepatitis, particularly for HBsAg, has been changing over time (Tables 4 and 5) [33]. Importantly, we drew from the entire population of blood-donors and also systematically reviewed the entire peer-reviewed HBV and HCV literature from Zimbabwe to obtain our prevalence estimates. Therefore, to the best of our knowledge, these data (shown in Tables 2 and 3 and Figures 4 and 5) are robust estimates of HBV and HCV biomarkers prevalence in blood-donors and the general Zimbabwean population, to date.

It is notable and worthy of further study, however, that despite high rates of heterosexually transmitted HIV infection, corresponding increase in HCV infections have not encountered or reported in Zimbabwe as has been observed in the US communities or southern European states with high heterosexually transmitted HIV infections [19,43,44]. Transmission patterns and acquisition for HBV in sub-Sharan African communities is dominantly in childhood and

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perhaps includes mother-to-child vertical transmission [12]. Although screening for chronic HBV is now undertaken in HIV treatment programs [31,40], it is still estimated that 95% of person with chronic HBV or HCV infection are unaware of their infection, which explains dearth of knowledge on the disease epidemiology in Zimbabwe [1,2]. Incorporation of full antenatal HBsAg screening and targeted HBV vaccination including the birth dose would fill an informational gap and also improve outcomes. Additional studies, particularly in children, that examine transmission patterns for both HBV and HCV are needed to better inform public health policy in low-resource settings such as Zimbabwe.

We present two lines of qualitative and quantitative evidence to suggest that HCV prevalence is increasing, while overall HBV prevalence in decreasing is Zimbabwe. We still hypothesise that HBV vaccination could have contributed to the later pending further evidence. First, decreasing HBsAg prevalence trends blood-donors aged 16–20 years age-group, majority of whom would have received the HBV vaccine in childhood (Figure 2). Those trends were supported by meta-regression analysis in Table 5 and an increase in the GD/BD HBsAg prevalence ratio from 2.47 before HBV vaccine introduction to 6.35 after the HBV vaccine introduction. Together these data support a clear decrease in HBV prevalence in some population groups, which can be attributed to the HBV vaccine. Second, a report of ten-fold increase in anti-HCV antibody in blood-donors between 2004 and 2010 reported by Apata et al. [33] which we also saw in patients >41 year in blood-donors between 2015 and 2018 (Figure 1).

There are several strengths and few limitations in our study. First, we used WHO-recommended approaches and age-population weights from the WHO World Standard Population which, not only removes temporary effects of historical events such as economic misfortunes, wars and famine on population-age composition, but also allows cross-comparison for periods 2000 through 2025 [26]. Thus, the 2015/2018 ASPR of anti-HCV antibody and HBsAg biomarkers from blood-donors are benchmarked for further comparison. Second, the systematic review and meta-analysis was exhaustive and supported trends observed with blood-donor data analysis. Second, we introduced GP/BD HBsAg prevalence ratio as a tool to track viral infections in the general population using routine surveillance data. The first limitation is that our biomarkers, particularly the HCV biomarker, do not indicate levels of viremia and they do not distinguish past infection from resolved infection [1,2]. Third, our aggregated blood-donor data did not distinguish first-time donors from repeat donors; hence, we could not separate viral incidence from prevalence. Incidence trends are better dynamic measures of disease burden.

Conclusions

Consistent with the introduction of the HBV vaccination, HBV prevalence is falling in Zimbabwe. Despite the apparent low HCV prevalence among blood-donors, temporal trends among those donors and point prevalent estimates in the general population suggest an increase in HCV burden in Zimbabwe. There is adequate reason for NBSZ to consider revision of currently used testing assays and screening protocols to benchmark them with those proven to carry the utmost minimal risk to reduce HCV and HBV transmission. Our data support expansion of HBsAg screening and increase in vaccination coverage to include

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key populations, such as HIV-infected persons with robust immune responses (CD4 counts >200) and pregnant women.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data availability statements

Data inputs and analytic code available upon request to the corresponding authors (MM) or (JGP).

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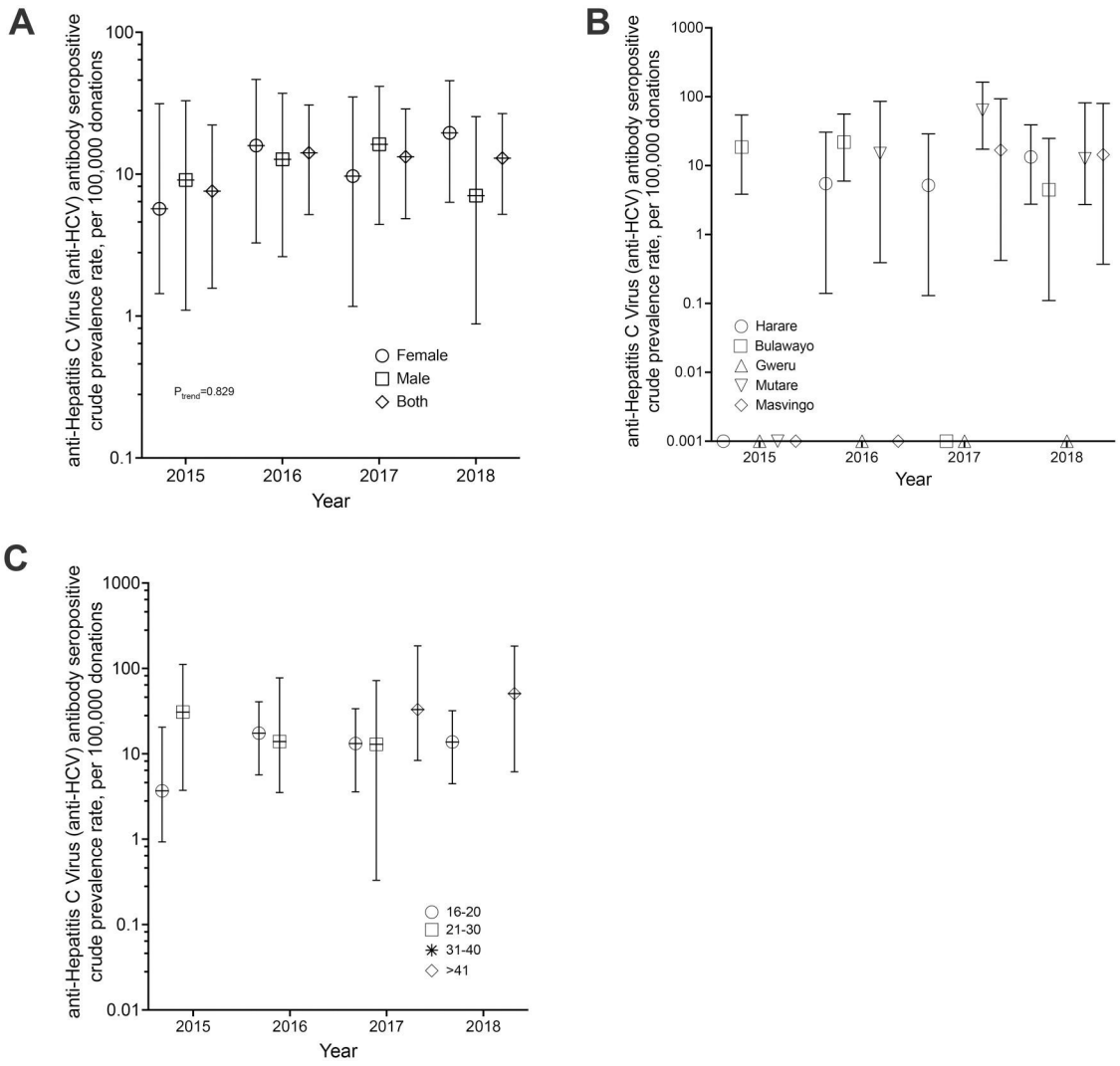


Figure 1.

Temporal trends and prevalence estimates of HCV (anti-HCV) biomarker. (A-C) shows the crude prevalence of anti-hepatitis C virus (anti-HCV) per 100,000 donors for the years 2015 through 2018 stratified by sex (A), by city (B) and by the age-groups (C). The y-axis is on a log scale and in some years some cities did not reported any cases. As shown there was not a significant trend demonstrated. Together, the data suggests that the overall increase in anti-HCV increase is driven by increased prevalence in the >41 years age group.

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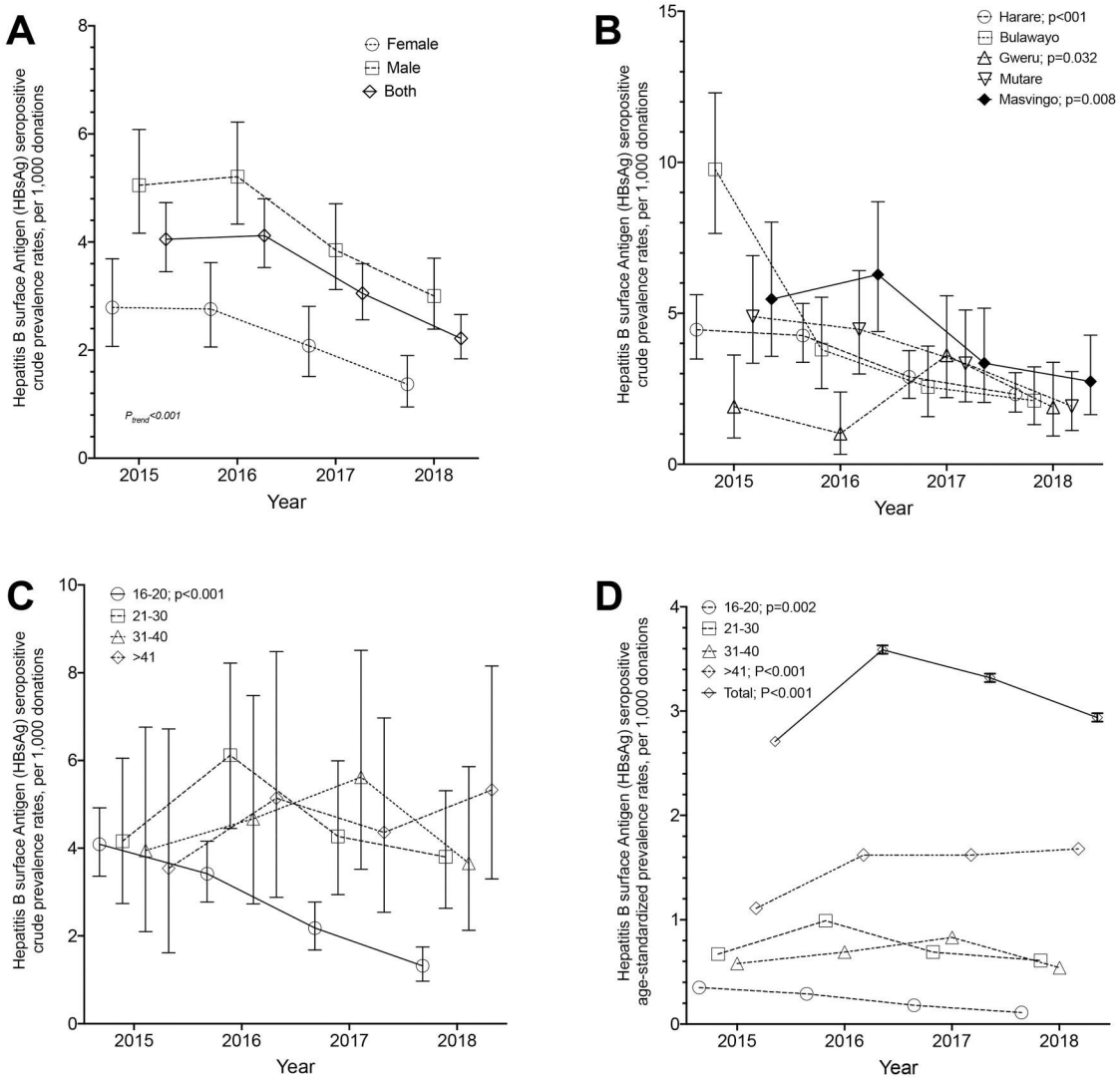


Figure 2.

Temporal trends and prevalence estimates of HBV (HBsAg) biomarker. (A–C) shows the crude prevalence of hepatitis B surface antigen (HBsAg) per 1000 donors for the years 2015 through 2018 stratified by sex (A), by city (B) and by the age-groups (C). As shown there is significant decrease in crude prevalence especially when the gender data is examined. (D) shows the age-standardized prevalence rate stratified by different age-groups and the total estimates (depicted with hollow diamond marker). There is persistent decline in age-standardized prevalence rate in the 16–20 year age-group, p =.002.

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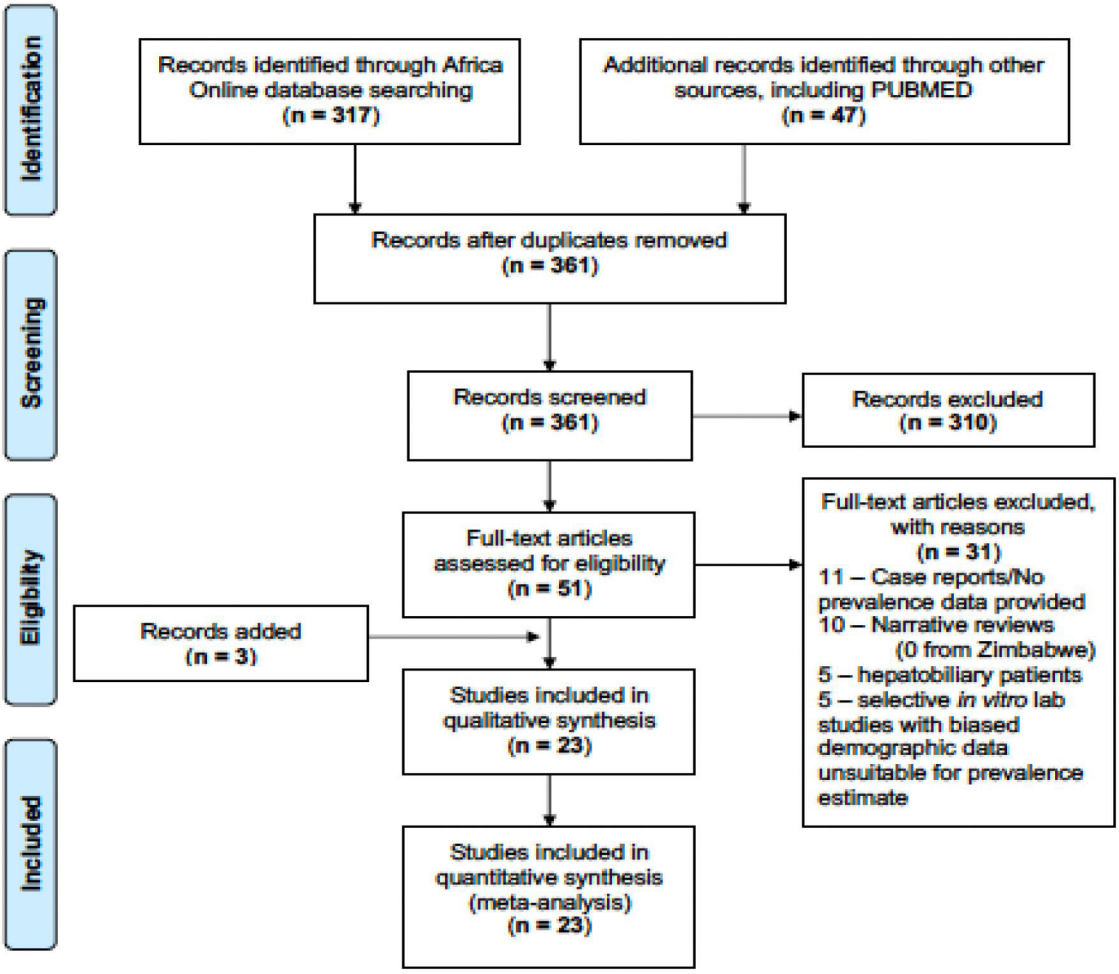


Figure 3.
PRISMA flowchart of article screening. Figure depicts the study selection steps followed.

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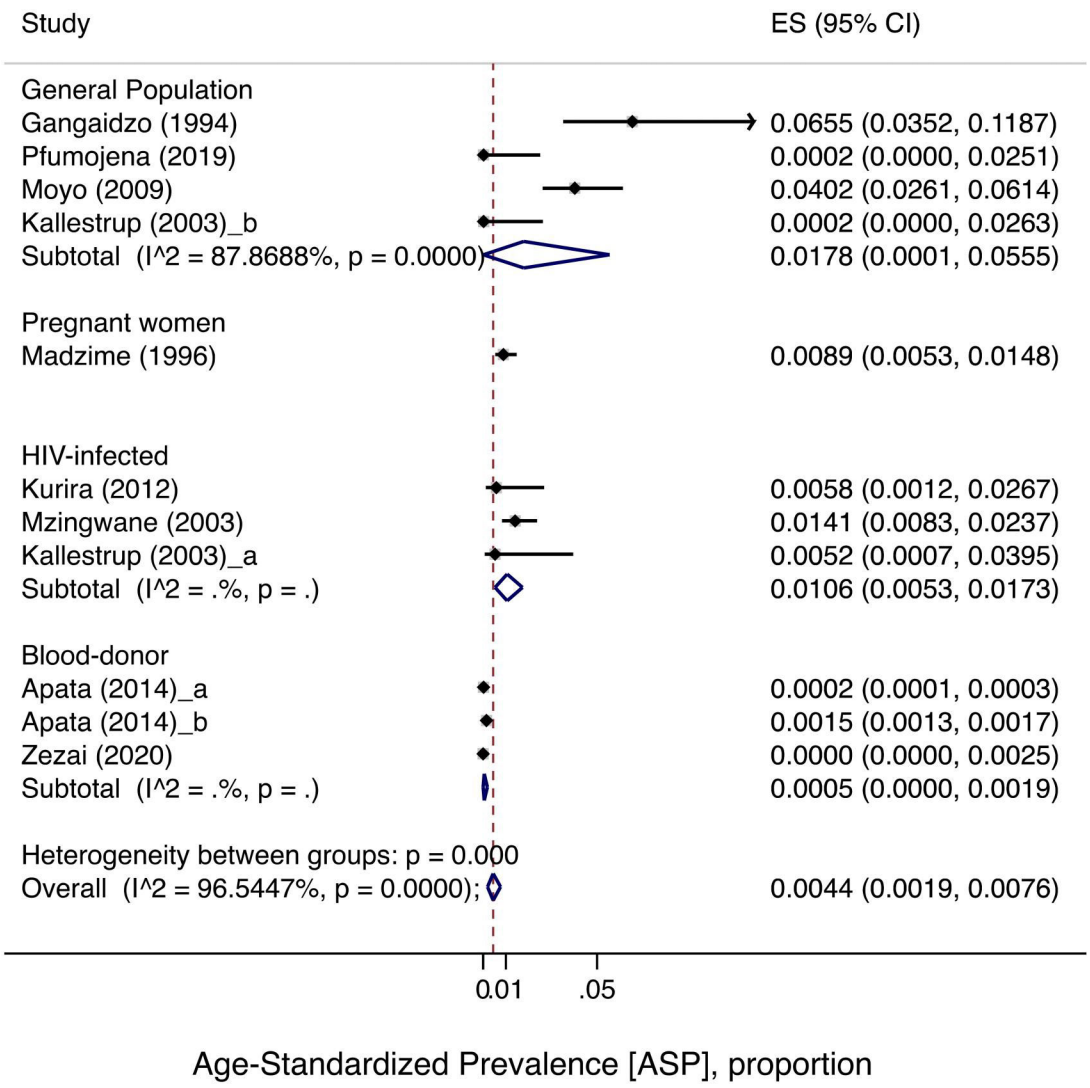


Figure 4.

Pooled estimates of age-standardized prevalence proportion of HCV biomarkers. Figure shows the age-standardized prevalence rate, as a proportion, stratified according to pooled estimates in the general public or population, pregnant women and in the blood-donor population groups. There was marked overall heterogeneity, I^2 = 965%; and the heterogeneity was also significant even within the selected groups.

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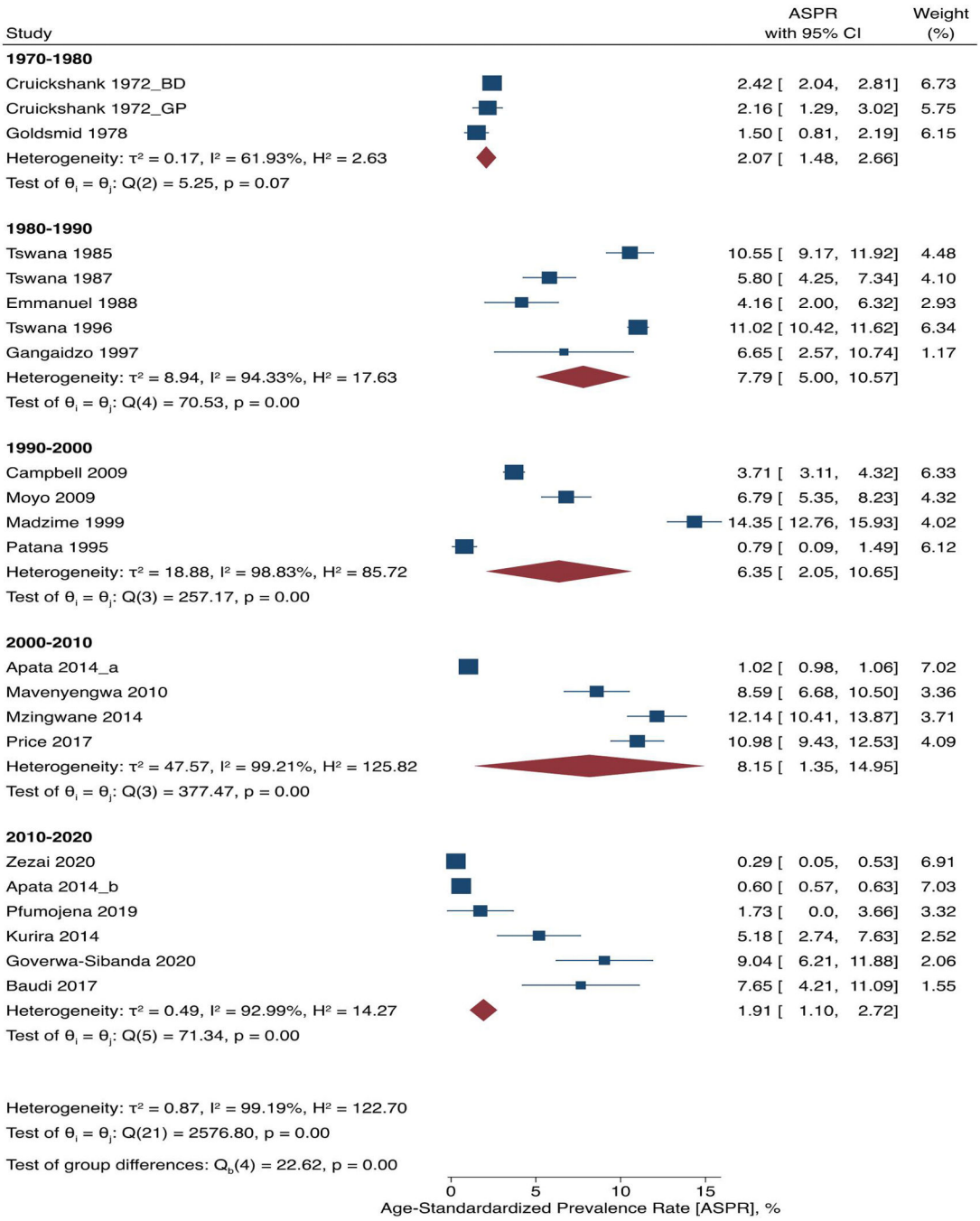


Figure 5.

Pooled estimates of age-standardized prevalence of HBV biomarkers. Figure shows the age-standardized prevalence rate, as a percent, stratified according to pooled estimates in the general public or population, for each decade from 1970–1980 to 2010–2020. Studies performed in blood-donors were combined with those done in the general public to estimate the weight general population prevalence in time. There was marked overall heterogeneity,

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I^2 = 99%; and the heterogeneity was also significant even within the selected groups, I^2 = 61.9%.

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Table 1.

Prevalence of anti-HCV seropositive and HCV indeterminate test results among blood donations: 2015–2018.

	Anti-HCV positive per 100,000 donors			Anti-HCV Indeterminate positive per 100,000 donors		
Variable	***n* = 22 (%)**	**CPR (95% CI)**	**ASPR (95% CI)**	***n* = 122 (%)**	**CPR (95% CI)**	**ASPR (95% CI)**
Sex						
Male	11 (50)	13.32 (6.65–23.84)	9.43 (4.71–16.88)	73 (60)	88.41 (69.30–111.2)	62.61 (49.08–78.75)
Female	11 (50)	11.16 (5.57–19.98)	7.90 (3.94–14.15)	49 (40)	49.73 (36.79–65.75)	35.22 (26.05–46.56)
Periods						
2015–2016	9 (41)	10.99 (5.02–20.86)	4.46 (2.57–6.35)	54 (44)	65.93 (49.53–86.02)	49.82 (41.40–58.24)
2017–2018	13 (59)	13.09 (6.97–22.39)	15.13 (8.22–22.04)	68 (56)	83.03 (64.47–105.3)	35.26 (29.10–41.42)
Age-groups						
16–20	15 (68)	12.25 (6.85–20.2)		80 (66)	65.32 (51.8–81.30)	
21–30	4 (18)	13.19 (3.59–33.78)		31 (25)	102.5 (69.46–145.1)	
31–40	0	0		5 (4)	32.29 (10.48–75.34)	
>41	3 (14)	23.63 (4.87–69.05)		6 (5)	47.26 (17.34–102.9)	
City						
Harare	5 (23)	6.58 (2.14–15.35)	4.66 (1.52–10.87)	73 (60)	96.01 (75.26–129.7)	67.99 (53.30–91.85)
Bulawayo	8 (36)	24.55 (10.06–48.36)	17.39 (7.12–34.25)	11 (9)	33.75 (16.85–60.39)	23.90 (11.93–42.77)
Gweru	0	0	0	9 (7)	42.95 (19.64–81.53)	30.42 (13.91–57.74)
Mutare	7 (32)	24.86 (10–51.23)	17.61 (7.44–36.28)	15 (12)	53.28 (29.82–87.87)	37.73 (21.12–62.23)
Masvingo	2 (9)	8.56 (1.04–30.91)	6.06 (0.74–21.88)	14 (12)	59.9 (32.75–100.5)	42.42 (23.19–71.17)

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Table 2.

Prevalence of HBsAg seropositive and HBsAg indeterminates among blood donations from 2015 to 2018.

	HBsAg positive per 1000 donors			HBsAg indeterminate positive per 1000 donors		
Variable	***n* = 593 (%)**	**CPR (95% CI)**	**ASPR (95% CI)**	***n* = 88 (%)**	**CPR (95% CI)**	**ASPR (95% CI)**
Sex						
Male	414 (70)	4.20 (3.80–462)	2.97 (2.69–3.27)	61 (69)	0.62 (0.47–0.79)	044 (0.33–0.56)
Female	179 (30)	2.17 (1.86–2.51)	1.54 (1.32–1.78)	27 (31)	0.33 (0.22–0.48)	0 23 (0.16–0.34)
Year Periods						
2015–2016	335 (56)	4.09 (3.66–4.55)	3.17 (3.00–3.34)	26 (30)	0.32 (0.21–0.46)	0.50 (0.32–0.68)
2017–2018	258 (44)	2.60 (2.29–2.94)	3.12 (2.61–3.62)	62 (70)	0.62 (0.48–0.80)	0.43 (0.36–0.50)
Age groups						
16–20	323 (54)	2.64 (2.36–2.94)		63 (72)	0.51 (0.40–0.66)	
21–30	138 (23)	4.55 (3.82–5.37)		11 (12)	0.36 (0.18–0.65)	
31–40	69 (12)	4.46 (3.47–5.64)		10 (11)	0.65 (0.31–1.19)	
>41	62 (11)	4.88 (3.74–6.26)		4 (5)	0.31 (0.09–0.81)	
City						
Harare	258 (44)	3.39 (2.99–3.83)	2.40 (2.12–2.71)	40 (45)	0.53 (0.38–0.72)	0.38 (0.27–0.51)
Bulawayo	90 (15)	2.76 (2.22–3.39)	1.95 (1.57–2.40)	12 (14)	0.37 (0.19–0.64)	0.26 (0.13–0.45)
Gweru	45 (8)	2.15 (1.57–2.87)	1.52 (1.11–2.03)	10 (11)	0.48 (0.23–0.88)	0.34 (0.16–0.62)
Mutare	99 (16)	3.51 (2.86–3.53)	2.49 (2.03–2.50)	11 (13)	0.39 (0.19–0.70)	0.28 (0.13–0.50)
Masvingo	101 (17)	4.32 (3.52–5.25)	3.06 (2.49–3.72)	15 (17)	0.64 (0.40–1.06)	0.45 (0.28–0.75)

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Table 3.

Characteristics of anti-HCV prevalence studies included in the systematic review and meta-analyses.

Author, Year, Ref	Study locale Enrolment Year Risk of Bias	Anti-HCV prevalence	Study aims, target population description, assays and main findings
Gangaidzo et al. (1997) [16]	Zaka, Masvingo Feb to May 1994 Bias - Moderate risk	<40 years – 1/64 >40 years – 10/78 Male – 5/72 Female – 6/70 Overall – 11/142	Study aim was to assess seropositivity to HCV in **150 healthy adults** from rural areas. Subjects were initially recruited to study genetic patterns of iron overload, hence some subjects were husband/wife pairs. However, index subjects were excluded from prevalence analysis to reduce risk of bias. Third generation enzyme immune-assays (EIA) used to test for anti-HCV antibodies (Abbott Labs). Anti-HCV CPR was 7.75% (95%CI, 3.87%–13.86%). Anti-HCV positives patients had significantly elevated liver function tests, suggesting ongoing disease. The ASPR was 6.65% (95% CI, 3.38%–11.54%). No children <12 years of age were enrolled.
Madzime et al. (2000) [17]	Harare Maternity Hospital, Harare June 1996 to Dec 1997 Bias – Minimum risk	<20 – 3/303 20–29 – 18/949 30–35 – 2/191 >35 – 2/134 Overall – 25/1591	Study aim was to assess HCV and syphilis seroprevalence in **1607 random pregnant women** about to deliver at Harare hospital, a tertiary maternity hospital. The hospital caters for 45% of deliveries in the City of Harare. Second/third generation EIA were used to test for HCV antibodies. Overall anti-HCV CPR was 1.6% (95% CI, 1.0%–2.2%) and 2.21% (95%CI, 0.81–4.74%) in those HBsAg positive. 25–29 years age-group, history of still births (4.29%) and prior syphilis infections (7.89%) were significantly associated with higher anti-HCV prevalence. The ASPR was 0.90% (95%CI, 0.59%–1.33%). No children <16years of age were enroled.
Kallestrup et al. (2003) [19]	Mupfure 2003 Bias – Minimum risk	Male – 0/45 Female – 1/224 Overall – 1/269	Retrospective study of anti-HCV prevalence in **269 patients** (124 HIV + versus 145 HIV−). Third generation enzyme immune-assay (Ortho HCV 3.0/Chiron RIBA) were used to test for HCV antibodies. Overall CPR was 0.37% (95%CI, 0.001%–2.05%) and there was no significant difference when compared by HIV status or gender. The ASPR 0.24% (95%CI, 0–1.35%). No children were enrolled in study.
Moyo et al. (2009) [34]	Harare & Bulawayo 1995 to 1996 Bias – Minimum	Male – 11/194 Females – 34/299 Overall – 45/493	Prospective study to measure population ferritin levels and its determinants in **493 individuals**; hence, HCV and HBV biomarkers were assessed of potential determinants. Third generation EIA (Biokit) were used to test for HCV antibodies. Overall CPR was 9.13% (95%CI, 6.74%–12.03%) and the CPR was significantly higher in females than males, relative risk = 0.499, $p = .037$. The ASPR 4.02% (95%CI, 2.96–5.29%). No children were enrolled in study.
Apata et al. (2014) [33]	Countrywide 2000/2004 2010/2011 Bias – Minimum risk	2000/2004 – 24/80,000 2010/2011 – 458/134,709 Overall – 482/214,709	Retrospective study to examine prevalence and trends in HCV biomarkers from **80,000 blood donations** seen between 2000 and 2004, and **134,706 blood donations** seen between 2010 and 2011, nationwide. Analysis is on aggregated data submitted to WHO. CPR increased eleven-folds from 0.03% (95%CI, 0.02%–0.05%) in 2000/2004 to 0.34% (95%CI, 0.31%–0.37%) in 2010/2011. The ASPR were 0.02% (95%CI, 0.01%–0.03%) and 0.22% (95%CI, 0.20%–0.24%), respectively.
Kurira (2014) [31]	Parirenytwa hospital, Harare Mar to July 2012 Bias – Minimum risk	>41 years – 2/68 Male – 2/89 Female – 0/139 Overall – 2/228	Study aim was to determine prevalence of HBV, HCV and coinfection of HBV/HCV in **228 HIV-infected patients** seen at a specialty clinic. Third generation ELISA (Diagnostic INC, CA, USA) were used to test for HBV and HCV biomarkers, including quantitative measures of viremia and genotypes. Anti-HCV CPR was 0.88% (95% CI, 0.11%–3.17%), while ASPR was 0.58% (95%CI, 0.07%–2.06%). HBV/HCV coinfection was not identified. 9/228 (4%) patients had prior blood transfusion, 2/9 (22%) of these were anti-HCV positive. No children were enrolled in study.
Mzingwane & Mamvura (2014) [24]	Harare 2003–2009 Bias – Minimum risk	HBsAg positive – 10/164 Overall – 19/957	A retrospective study of **957 HIV-infected patients** enrolled in HIV treatment program (TDF/3TC) to characterise HBV, HCV and coinfection with HIV. Abbott Murex and DiaSorin ETI-EBK PLUS EIA kits were used to measure anti-HBc biomarkers. Overall anti-HCV CPR was 1.99% (95%CI, 1.20%–3.08%) and 1% in those with active HBV. The ASPR was 1.41%(95%CI, 0.85%–2.18%). There were no children <18years of age enrolled in study.
Pfumojena et al. (2019) [32]	Harare Feb to May 2014 Bias – Minimum risk	Physicians – 0/48 Nurses – 0/104 Overall – 0/152	Study aim was to determine HCV prevalence and measure levels of viremia among **152 healthcare workers** at Parirenyatwa hospital; a tertiary teaching institution in Harare. One Step Novatest (Atlas Link, Beijing, China) was used to test for anti-HCV antibodies. HCV infection was not identified. No children enrolled in study.
Zezai et al. (2020) [9]	Country-wide Jan to Dec, 2018 Bias – Minimum risk	Male – 0/972 Female – 0/614 Overall – 0/1586	Study aim was to determine the HCV prevalence in **1586 routine blood-donors** first-time and repeat blood-donors in one year at 5 selected blood donation centres in Zimbabwe. ARCHITECT HBsAg Qualitative II, and Anti-HCV, from Abbott Labs tests used. HCV infection was not identified. No children <16years of age were enrolled in study.

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Table 4.

Characteristics of HBsAg studies included in the systematic review and meta-analyses.

Author, Year, Ref	Study locale Enrolment Year Risk of Bias	HBsAg Prevalence	Study aims, target population description, assays and main findings
A. Before HBV vaccination			
Cruickshank et al. (1972) [35]	Harare, Bulawayo, Nyanga, Kariba, Dande valley and, Zambezi Valley 1971–1972 Bias – Minimum risk	Blood-donors – 155/5489 Whites – 2/1275 Blacks – 153/4214 Police – 28/387 General pop – 26/1036	Cross-sectional prevalence study of Australia antigen (HBsAg) in **6525 healthy individuals** (5250 Blacks versus 1275 Whites), also divided as 5489 regular urban blood-donors versus 1036 rural healthy Black individuals, **including children and pregnant women**, selectively drawn from the general population. First generation EIA (cross-over immune-electrophoretic [CIEP] assay) using anti-Aul serum was used to test presents of HBsAg. Blood-donor HBsAg crude prevalence rate (CPR) 2.82% (95%CI, 2.40%–3.30%), while that in the contemporary general population was 2.51% (95%CI, 1.65%–3.66%), $p = .573$. HBsAg CPR was significantly higher in Black blood-donors than Whites ($p < .0001$), but there was no significant difference between Black blood-donors and the general population, $p = .075$. The study identified police forces, (28/387), as high risk groups among urban Blacks. HBsAg age-standardized prevalence rate (ASPR) in all blood-donors was 2.42% (95%CI, 2.06%–2.84%), while that in general population was 2.16% (95%CI, 1.42%–3.14%).
Goldsmid & Hurst (1978) [36]	Country-wide, Army Medical Corps Bias – Minimum risk	Blacks – 20/262 Non-Blacks – 0/564 Overall – 20/826	Cross-sectional prevalence study of HBsAg in **826 soldiers** enlisted in the Rhodesian army. First generation EIA, the CEIP assay was used and compared to the Hepatest. HBsAg prevalence was higher 12.5% versus 7.6%, in Blacks, using the Hepatest. HBsAg CPR in Blacks was 7.63%(95%CI, 4.72%–11.54%), while the overall CPR was 2.42% (95%CI, 1.49%–3.71%). Overall ASPR was 1.50% (95%CI, 0.92%–2.30%).
Tswana (1985) [13]	Country-wide: Gweru, Wedza, Kariba, Kadoma, Masvingo Bias-Minimum risk	Female – 74/717 Male – 127/754 Overall – 201/1471	Cross-sectional study of HBsAg prevalence in **1471 random individuals**, ages 11–70 years, drawn from the general healthy population; and to compare the proportion across different age-groups. ELISA, with confirmatory Raphadex assays (Ortho Diagnostics) were used. HBsAg CPR of males was significantly higher than females by 6.52%(95%CI, 3.2%–10.0%), $p = .0003$. Overall HBsAg CPR was 13.66%(95%CI, 11.95%–15.52%), while the ASPR was 10.55% (95%CI, 9.23–11.98%). Masvingo had 19.7% and Kadoma 17.2%.
Tswana & Berejena (1987) [14]	Harare 1986–1987 Bias – Minimum risk	Dental worker – 6/100 General pop – 47/500 Blacks – 50/505 Whites – 3/95 Overall – 53/600	Cross-sectional study of HBeAg and HBsAg prevalence in **600 persons participating at dental practices** (100 dental workers and 500 general population) in different races (White versus Blacks). ELISA Abbott assays and platforms were used. HBsAg CPR of 6% (95%CI, 2.23%–12·60%) in dental workers compared to 9.4% (95%CI, 6.99%–12.30), $p = .273$ from the general population. Overall CPR was 8.83% (95%CI 6.69%–11.40%), the corresponding ASPR was 5.80% (95%CI, 4.39%–7.48%).
Emmanuel et al. (1988) [21]	Harare Bias - Moderate risk	HW – 15/226	Study aim was to compare the prevalence of HBV markers in **226 hospital workers (HW) to 97 random volunteer blood-donors**. Abbott lab kits were used to test HBV markers and confirmed using AUSRIA 11–125 radioimmunoassay. HBsAg CPR was 6.64% (95%CI, 3.76%–10.71%). The HBsAg ASPR was 4.10% (95%CI, 2.36%–6.68%). HBV biomarkers not separated among blood-donors by the individual biomarkers. However, the authors report that *"HBV risk in blood-donors was similar to HW"*.
Patana et al. (1995) [23]	Chiweshe 1995 Bias - Minimum	Overall – 6/299	Study aim was to examine prevalence of HBV and schistosomiasis markers in **299 pregnant women**. HBsAg CPR was 2.01% (95%CI, 0.74%–4.32%) while the ASPR was 0.79% (95%CI, 0.29%–1.70%)
Moyo et al. (2009) [34]	Harare & Bulawayo 1995–1996 Bias – Minimum risk	Male – 45/194 Female – 31/299 Overall – 76/493	Prospective study to measure population ferritin levels and its determinants in **493 individuals**; hence, HCV and HBV biomarkers were assessed of potential determinants. Third generation EIA (Biokit) were used to test for HCV antibodies. Overall HBsAg CPR was 15.41% (95%CI, 12.34%–18.91%), while the ASPR was 6.79% (95%CI, 5.43%–8.32%).
Tswana et al. (1996) [15]	Country-wide, all 9 provinces April 1989 to December 1991 Bias – Minimum risk	Overall – 523/3394	Study aim was to estimate prevalence of hepatitis biomarkers [HBsAg, HBeAg, Anti-HBs, Anti-HBc, Anti-HBe] in **3394 random healthy individuals** drawn from the general Zimbabwean population. ELISA assay (Abbott labs) used. Overall HBsAg CPR was 15.41% (95% CI, 14.21%–16.7%) [males 16.8%, females 15.4%], while 25% (95% CI, 20.85%–28.39%) of these were HBeAg positive. Overall ASPR was 11.02% (95%CI, 10.42%–11.62%). ASPR

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Author, Year, Ref	Study locale Enrolment Year Risk of Bias	HBsAg Prevalence	Study aims, target population description, assays and main findings
			was 10.64% (95%CI, 9.50%–11.86%) in females, which was significantly lower than 12.50% (95%CI, 11.10%–14.06%) in males, $p = .045$.
Gangaidzo et al. (1997) [16]	Zaka, Masvingo Feb to May 1994 Bias – Moderate	<40 years – 15/64 >40 years – 5/78 Overall – 11/142	Study aim was to estimate prevalence of HBsAg in **150 healthy adults** aged 12–84 years from rural Zimbabwe. CPR HBsAg prevalence was 14.08% (95%CI, 8.60%–21.75%), while the ASPR was 6.65% (95% CI, 3.38%–11.54%). None of the study participants with HBsAg had elevated liver enzymes.
B. After HBV vaccination			
Madzime et al. (1999) [20]	Harare Maternity Hospital, Harare June 1996 to June 1997 Bias - Minimum risk	<20 years – 50/217 20–29 year – 154/583 30–35 years – 22/101 >35 years – 18/72 Overall – 246/984	HBsAg carrier and infectivity prevalence study of **1,000 random pregnant women** about to deliver at a tertiary referral maternity hospital in Harare. The hospital caters for 45% of deliveries in the City of Harare. EIA was used to test for HBsAg in blood, while HBeAg was tested in all HBsAg positive samples. CPR HBsAg prevalence was 25% (95%CI, 22%–28%), while ASPR was 14.35% (95%CI, 12.81%–15.97%). 8/246 (3.25%) samples were HBeAg positive which meant that only 0·81% of women were at an elevated risk of transmitting HBV to their new-born babies.
Price et al. (2017) [37]	Harare 2006 Bias - Minimum	Overall – 167/998 HBeAg – 52/150	Study aim was to determine HBV markers, including HBV DNA levels in **998 HIV-infected patients** in the Zimbabwe DART trial. Third general COBAS EIA used to measure HBV markers. HBsAg CPR was 16.73 (95%CI, 14.47–19.19%), while ASPR was 10.98% (95%CI, 9.50%–12.60%). HBeAg was 35%.
Campbell et al. (2009) [38]	Harare March 1993 to June 1997 Bias - Minimum	Overall – 144/2750	The Zimbabwe AIDS Prevention Project (ZAPP) was a longitudinal cohort study of **3383 men and 373 wives** recruited at 40 factories in Harare. Study participants, ages 17–75 years, were assessed regularly for sexual risk factors and also tested for HHV-8, HIV, HBsAg and syphilis. Increase in HIV prevalence was independently significantly associated with HBsAg prevalence among other factors, such as age and paid sex. HBsAg CPR was 5.24%(95%CI, 4.43%–6.14%), while the ASPR was 3.71% (95%CI, 3.14%–4.35%).
Mavenyengwa et al. (2010) [22]	Harare, Guruve, Chisungo, January 2003 to December 2005 Bias - Minimum	Harare – 15/189 Chisungo – 15/60 Guruve – 39/169 Overall – 69/418	To compare prevalence of HBV, HIV and Group B Streptococcus con-infection in **418 pregnant women**. HbsAg, anti-HBc and anti-HBs were examined on third generation EIA Abbott platform. Overall HBsAg CPR was 16.51% (95%CI, 13.08%-20.42%), while APR was 8.59% (95%CI, 6.80%-10.62%).
Kurira, P (2014) [31]	Parirenyatwa hospital, Harare March to July 2012 Bias - Minimum	Male – 13/89 Female – 5/139 Overall – 18/228	Study aim was to determine prevalence of HBV, HCV and coinfection of HBV/HCV in **228 adult HIV-infected patients** seen at a specialty clinic at Parirenyatwa: a tertiary teaching hospital in Harare. ELISA (Diagnostic INC, CA, USA) were used to test for HBV and HCV biomarkers. HBsAg CPR was 7.89% (95%CI, 4.68%-12.48%), while the ASPR was 5.18% (95%CI, 3.12%-8.0%). HBV/HCV coinfection was not identified. None of the 9 transfused patients were HBsAg positive.
Mzingwane, ML & Mamvura, T (2014) [24]	Harare 2003 to 2009 Bias – Minimum	Male – 77/405 Female – 87/552 Overall – 164/957	A retrospective study of **957 HIV-infected patients** enrolled in HIV treatment program and to characterise HBV, HCV and coinfection with HIV. Third generation EIA Abbott Murex assays were used to measure HBsAg, HBeAg, anti-HBc, and anti-HCV biomarkers. HBsAg CPR prevalence was 17.14% (95%CI, 14.61%–19.70%), and HBeAg was observed in 103 (10.8%) of those with HBsAg positive. The ASPR was 12.14% (95%CI, 10.48%–13.94%).
Apata, IW et al. (2014) [33]	Countrywide 2000/2004 2010/2011 Bias – Minimum risk	2000/2004 – 2,340/150,000 2010/2011 – 1239/134,709 Overall – 3479/284,709	Retrospective study to examine prevalence and trends in HBV biomarkers from **150,000 blood donations** seen between 2000 and 2004, and **134,706 blood donations** seen between 2010 and 2011, nationwide. Analysis is on aggregated data submitted to WHO. CPR decreased 40% from 1.56% (95%CI, 1.50%–1.62%) in 2000/2004 to 0.92% (95%CI, 0.87%–0.97%) in 2010/2011. The ASPR were 1.02 (95%CI, 0.98%–1.06%) and 0.60% (95%CI, 0.57–0.64%), respectively.
Baudi et al. (2017) [39]	Harare June to September 2014 Bias - Moderate	Overall – 16/176	The study objective was to determine prevalence of HBV genotypes in **176 HIV co-infected**. Third generation chemiluminescence assays used to test HBsAg and HBV DNA levels measured. HBsAg CPR was 10.80% (95% 6.62%–16·34%), while ASPR was 7.65% (95%CI, 4.69%–11.57%). All isolates were A1 genotypes and HBV DNA was detectable in 12/16 (75%).

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Author, Year, Ref	Study locale Enrolment Year Risk of Bias	HBsAg Prevalence	Study aims, target population description, assays and main findings
Pfumojena et al. (2019) [32]	Harare February to May 2014 Bias – Minimum risk	Physicians – 1/48 Nurses – 3/104 Overall – 4/152	Study aim was to determine HBV prevalence, vaccination status and describe susceptibility patterns among **152 healthcare workers (HCW)** at a tertiary teaching institution. One Step Novatest (Prechek Bio, Anaheim, CA, USA) was used for both serology and antigen tests. HBsAg CPR prevalence was 2.6% (95%CI, 1.0%–6.6%), while ASPR was 1.7% (95%CI, 0.5%–4.3%). None of the HCW with chronic HBV infection were put on treatment!
Goverwa-Sibanda et al. (2020) [40]	Bulawayo October 2017 to April 2019 Bias - Minimum risk	5–14 years – 2/33 15–24 years – 5/62 25–49 years – 28/228 >50 years – 3/38 Overall – 38/361	Study aim was to determine prevalence of HBV and coinfection of HBV/HCV in **422 HIV-infected children and adult patients** seen at a specialty clinic at Mpilo: a tertiary teaching hospital in Bulawayo. This is the first peer-reviewed study that examined HBV in HIV-infected children from Zimbabwe, revealing prevalence of 6.06%(95%CI, 0.74%–20.23%) in those between the ages of 5–14years. The overall prevalence of 10.53% (95%CI, 7.56%–14.16%) is significantly lower by mean of 6.61% (95% 2.25%–10.97%), $p = .003$, when compared to an earlier Harare study see ref [24]. The ASPR was 9.04% (95%CI, 6.49%–12.17%).
Zezai et al. (2020) [9]	Countrywide Jan to Dec, 2018 Bias - Minimum risk of bias	Overall – 7/1586	Study aim was to determine the HCV prevalence in **1586 routine blood-donors** [232 first-time and 1354 repeat blood-donors] in one year at 5 selected blood donation centres in Zimbabwe. ARCHITECT HBsAg Qualitative II, and Anti-HCV, from Abbott Labs tests used. The CPR was 4.41% (95% CI, 1.77–9.09), while the APR was 2.90% (95% CI, 1.16 – 5.97).

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Table 5.

Meta-regression analysis of the joint factors associated with the seroprevalence variance of hepatitis B surface antigen (HBsAg) biomarkers before and after hepatitis B vaccine introduction in Zimbabwe.

Variable	Coefficient	95% Confidence Interval	*p*-value
Before HBV Vaccine introduction Meta-regression model. R^2 = 52.47%			
1970–1980 decade		-Referent-	
Subsequent decades	6.00	0.18–11.01	.045
Constant	25.82	6.37–45.28	.019
After HBV Vaccine introduction Meta-regression model. R^2 = 78 09%			
General Population		-Referent-	
HIV-infected persons/Pregnant Women	7.80	2.51–13.10	.015
2000–2010 decade		-Referent-	
Subsequent decades	−2.45	−5.37 to 0.47	.080
Constant	12.85	8.41–17.28	.001

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